DO NOW — RECALL

Appendix 5d OCR notation challenge

Name: _____ Class: _____ Date: _____

1. Why can OCR array[5] access index4 but not index5?

2. What is byRef?

3. What does readLine return?

4. In C# array [2,3,5,7], what is a[2] and total sum?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Five items are indexed 0–4.

2. Model answer

Parameter passing allowing modification of caller's underlying variable.

3. Model answer

Next line of text from a file.

4. Model answer

a[2]=5; total=17.

By the end, you can…

• Use exact indexing

• Trace subroutines and parameters

• Read file operations

Vocabulary — know these terms

array — Fixed-length zero-indexed sequence.

byVal — Pass a value/copy.

byRef — Pass a reference to original.

readLine — Read the next file line.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check reasoning

After addOne(n), n=7 and print displays 7.

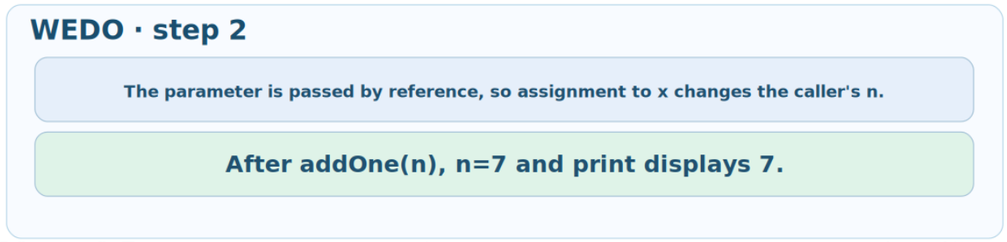
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a function and a procedure that changes an argument by reference.

The parameter is passed by reference, so assignment to x changes the caller's n.



WEDO · step 2: The parameter is passed by reference, so assignment to x changes the caller's n.; After addOne(n), n=7 and print displays 7.

NEW LEARNING

Read, annotate and explain

Use exact indexing

OCR arrays use zero-based indexing; the number in array declaration is the number of elements.

Trace subroutines and parameters

A procedure does not produce a return value; a function returns a result. byVal passes a copy; byRef allows the original parameter to change.

Read file operations

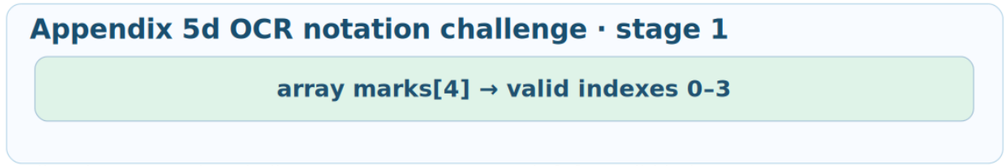
openRead, readLine, endOfFile and close follow the OCR Appendix 5d conventions. Always close an opened file.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



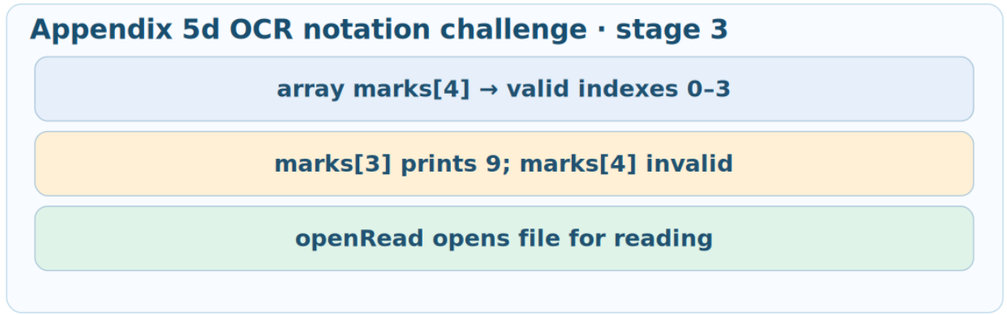
Appendix 5d OCR notation challenge · stage 1: array marks[4] → valid indexes 0–3

New Learning



Appendix 5d OCR notation challenge · stage 2: array marks[4] → valid indexes 0–3; marks[3] prints 9; marks[4] invalid

New Learning



Appendix 5d OCR notation challenge · stage 3: array marks[4] → valid indexes 0–3; marks[3] prints 9; marks[4] invalid; openRead opens file for reading

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

array values[4] can validly access values[4].

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

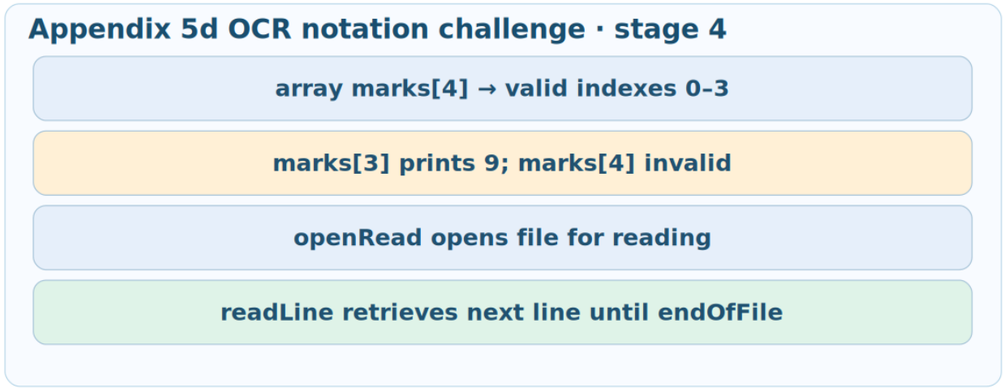
Question 3

Compare procedure addTwo(x:byVal) with addTwo(x:byRef), each executing x=x+2, for caller n=5. [6]

New Learning · original lesson diagrams

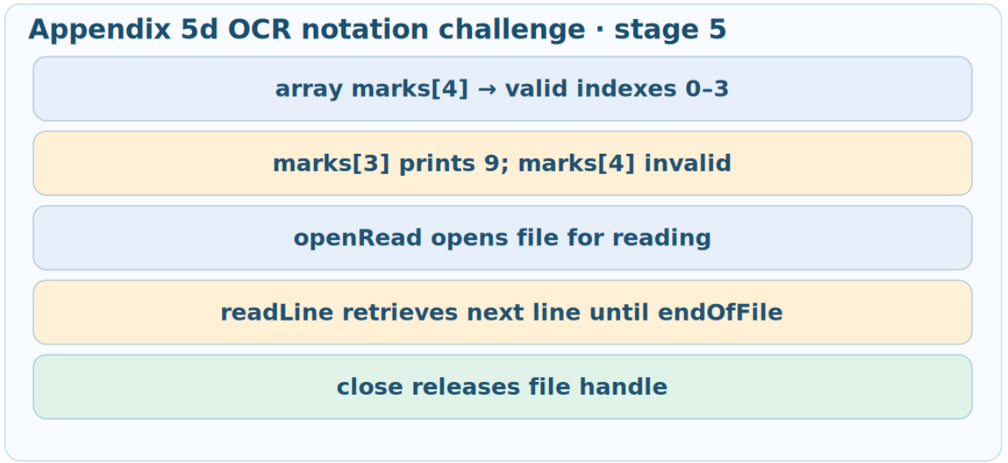
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Appendix 5d OCR notation challenge · stage 4: array marks[4] → valid indexes 0–3; marks[3] prints 9; marks[4] invalid; openRead opens file for reading; readLine retrieves next line until endOfFile

New Learning



Appendix 5d OCR notation challenge · stage 5: array marks[4] → valid indexes 0–3; marks[3] prints 9; marks[4] invalid; openRead opens file for reading; readLine retrieves next line until endOfFile; close releases file handle

I DO — WORKED EXAMPLE

Follow the worked method

Trace a 3-element OCR array and identify the valid indexes.

Predict first

Array declaration creates three positions: indexes 0,1,2.

Check reasoning

scores[1] holds 4, so output is 4.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace the OCR reference-language array below. State the output and explain why accessing a[3] would be invalid. [5]
CODE:
array a[3]
a[0] = 4
a[1] = 8
a[2] = 12
print(a[2])

Question 2

Write OCR exam reference language to read and print every line of the file scores.txt, and then close it. [7]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The parameter is passed by reference, so assignment to x changes the caller's n.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a 3-element OCR array and identify the valid indexes.

Array declaration creates three positions: indexes 0,1,2.



IDO · step 1: Array declaration creates three positions: indexes 0,1,2.

Trace a 3-element OCR array and identify the valid indexes.

Array declaration creates three positions: indexes 0,1,2.



IDO · step 2: Array declaration creates three positions: indexes 0,1,2.; scores[1] holds 4, so output is 4.

WE DO — GUIDED PRACTICE

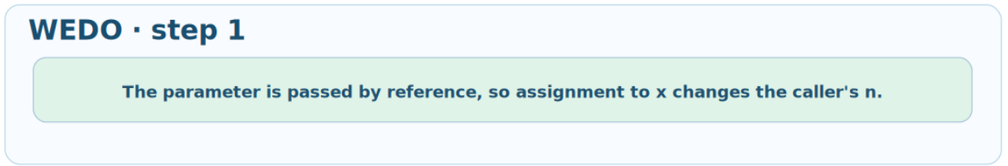
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a function and a procedure that changes an argument by reference.

The parameter is passed by reference, so assignment to x changes the caller's n.



WEDO · step 1: The parameter is passed by reference, so assignment to x changes the caller's n.